

Quantization-Aware Digital PID Control of an Inverting Buck–Boost Converter on Two Microcontrollers (ESP32-S3 and STM32F103): Measured Latency, Effective-Resolution, and Adaptive Update Management

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ABSTRACT Digital control of switching power converters is shaped by three implementation-quantized quantities: the ADC resolution of the sensed output, the DPWM resolution of the commanded duty, and the sampling period at which controller and power stage are synchronized. Whether the resulting design limits are properties of the hardware or of the software stack that implements the loop is rarely asked. This paper asks that question for an inverting buck-boost converter ($V_{\text{ref}} = -12\text{ V}$, $f_{\text{sw}} = 100\text{ kHz}$). A 79-run switched-model study first locates the critical sampling boundary near $3.5\text{--}4 \times T_{\text{sw}}$ and shows the three quantizers interact super-additively. The same controller is then ported, from one shared firmware source, to two microcontrollers: an ESP32-S3 (240 MHz, hardware FPU) and an STM32F103 Blue Pill (72 MHz, no FPU). The central finding is an abstraction-layer tax: through the Arduino API the STM32F103 is as latency-bound as the ESP32-S3 (ADC $\approx 73\text{ }\mu\text{s}$, PWM $\approx 24\text{ }\mu\text{s}$, and a $232\text{ }\mu\text{s}$ soft-float quantizer), but bare-register access recovers a $4.2\text{ }\mu\text{s}$ ADC and a $26\text{ }\mu\text{s}$ paced-loop floor ($2.6 \times T_{\text{sw}}$) that crosses the simulated sampling boundary, which the ESP32-S3 path ($73\text{ }\mu\text{s}$ floor) cannot reach. A boundary-pacing experiment validates the achievable schedule; closed-loop regulation across the boundary awaits a real LC plant, which is wired in software and deferred to hardware validation. An adaptive controller that switches update schedules by a measured error indicator cuts update frequency by 84% on the ADC-bound ESP32-S3 while holding steady-state error within $\pm 40\text{ mV}$. All simulation and measurement data are publicly available, and the complete firmware and analysis pipeline accompany the paper.

INDEX TERMS digital PID control, quantization, buck-boost converter, ESP32-S3, STM32F103, cross-platform, adaptive sampling, measurement latency

I. INTRODUCTION

Digital control loops are now standard in switching power converters, driven by programmability, immunity to analog component drift, and integration with digital loads [1], [2]. The complete digital loop contains two unavoidable quantizers: the analog-to-digital converter (ADC) that samples the output voltage, and the digital pulse-width modulator (DPWM) that resolves the duty ratio. Their resolutions, together with the sampling period of the controller, jointly determine regulation quality. Classic work established that coarse quantiza-

tion causes steady-state limit cycling and conditions between DPWM and ADC resolution for its elimination [3], [4], [5], [6], [7], while quantization-effect modeling permits quantitative prediction of these oscillations [8]. Recent work extends the analysis to MHz-range GaN converters using dither and sigma-delta modulation [9], [10], [11], [12] and to design guidelines for two-loop current-mode digital controllers [10].

Digital control relieves the analog constraints of passive compensation, enables programmability and auto-tuning, and integrates naturally with digital

loads [1], [2]. In exchange, the designer accepts three implementation-quantized quantities that an analog loop never sees: the ADC resolution of the measured output, the DPWM resolution of the commanded duty, and the sampling period at which controller and power stage are synchronized; DPWM-architecture and delay considerations are central to achievable resolution [13], [14], [15], [16].

The theoretical basis is well established. Peterchev and Sanders derived conditions on ADC/DPWM resolution trade-offs for limit-cycle elimination and introduced digital dither to recover effective DPWM resolution [4]. Peng, Prodić, Alarcón, and Maksimović derived static and dynamic quantization models explaining the origins of limit-cycle oscillations [5], and Maksimović and Zane provided the small-signal discrete-time framework used to design digitally controlled converters [8]. A fuller survey appears in [17]. On the application side, low-resolution non-uniform ADC quantization has been shown to improve dynamic response [18], and the full control loop benefits from predictive structures [19], [20], [21], [22], [23].

Two observations motivate this work. First, most quantization studies isolate a single stage, whereas a practical controller is constrained in all three quantities simultaneously; the interaction between sampling period, ADC word length, and PWM word length is rarely treated as one design problem. Second, the buck-boost converter, whose inverted output and non-minimum phase control path [24] make regulation inherently harder, is underrepresented in the digital-control literature: most buck-boost-specific digital studies consider time-domain or PID structures at a single resolution [25], [26], [27], [28], [29] and do not sweep the full implementation space. The classical converter model itself is textbook material [30].

The central question of this work is therefore: how do sampling period, ADC quantization, PWM resolution, and real-time computational latency jointly affect the closed-loop performance of a microcontroller-controlled inverting buck-boost converter, and can a lightweight quantization-aware adaptive PID maintain acceptable regulation while reducing unnecessary controller-update effort — and do the answers depend on the microcontroller? Answering it requires three levels of validation (an approach related to event-triggered and self-triggered control [31], and to adaptive-sampling converter control [32], [33]) that earlier studies treat separately: a continuous (analog) baseline, a digital simulation that introduces sampling, ADC and PWM quantization and computational delay, and closed-loop measurements on two embedded platforms. The microcontrollers are not merely devices that run the PID; they are the experimental vehicles for exposing quantities a simulator cannot produce directly: the acquisition time of the on-chip ADC, the execution time of the control step

(with and without a hardware FPU), the update latency of the PWM peripheral, and the resulting control-loop latency and CPU utilization. The study uses one shared firmware source on an ESP32-S3 (240 MHz, Arduino core, LEDC + `analogRead`) and an STM32F103 Blue Pill (72 MHz, no FPU, STM32duino), whose $17\times$ bare-register ADC advantage isolates whether the predicted sampling boundary is a hardware fact or a firmware artifact.

This paper reports a simulation-to-hardware investigation of a digitally controlled inverting buck-boost converter under simultaneous quantization of sampling period, ADC resolution, and PWM resolution. Contributions: (1) a 79-run factorial study that separates the individual effect of each quantizer from their combined effect, locates the critical sampling boundary near $3.5\text{--}4\times T_{\text{sw}}$, and quantifies the super-additive interaction between the three stages; (2) an objective worst-case selection made after the sweep, not a priori; (3) the identification of a half-scale ADC coincidence ($V_{\text{ref}} = -\text{FS}/2$) that masks steady-state error at low ADC resolutions; (4) an ESP32-S3 closed-loop implementation whose control-loop latency is measured and decomposed into ADC acquisition, PID execution, and PWM update; (5) a cross-platform replication on an STM32F103 that measures the same decomposition through both the Arduino abstraction layer (ADC $72.8\text{ }\mu\text{s}$, PWM $23.9\text{ }\mu\text{s}$) and bare-register access (ADC $4.23\text{ }\mu\text{s}$, PWM $5.88\text{ }\mu\text{s}$), revealing a $17\times$ ADC tax imposed by the abstraction layer and a $232\text{ }\mu\text{s}$ soft-float quantizer cost that vanishes with fixed-point or a hardware FPU; (6) a bare-register sampling-boundary sweep (mode T) on the STM32F103 that paces the loop at every T_s/T_{sw} rung from $1\times$ to $5\times$ and crosses the predicted boundary while the ESP32-S3 path cannot reach it (floor $73\text{ }\mu\text{s} \approx 7.3\times T_{\text{sw}}$ vs $26\text{ }\mu\text{s} \approx 2.6\times T_{\text{sw}}$); (7) an effective-resolution experiment on the ESP32-S3 showing that the measured acquisition noise floor masks the quantization effects the simulation predicts; and (8) a lightweight quantization-aware adaptive PID that switches update effort by a measured period and ripple indicator and validates the predicted performance-versus-resource trade-off with an update-reduction ratio of roughly 84% on the ESP32-S3. The novelty lies not in variable-rate sampling or dithering individually, but in their integration into a resource-aware control framework that is characterized, implemented, and experimentally validated across two microcontrollers from one shared firmware source. Hardware closed-loop validation with a real LC plant is software-ready and deferred to the next revision.

II. SYSTEM MODEL AND DIGITAL CONTROL

The plant is an ideal inverting buck-boost converter (Fig. 1) with input $V_{\text{in}} = +12\text{ V}$, reference $V_{\text{ref}} = -12\text{ V}$, $L = 100\text{ }\mu\text{H}$, $C = 220\text{ }\mu\text{F}$, switching frequency $f_{\text{sw}} = 100\text{ kHz}$ ($T_{\text{sw}} = 10\text{ }\mu\text{s}$), and load $R = 10\text{ }\Omega$.

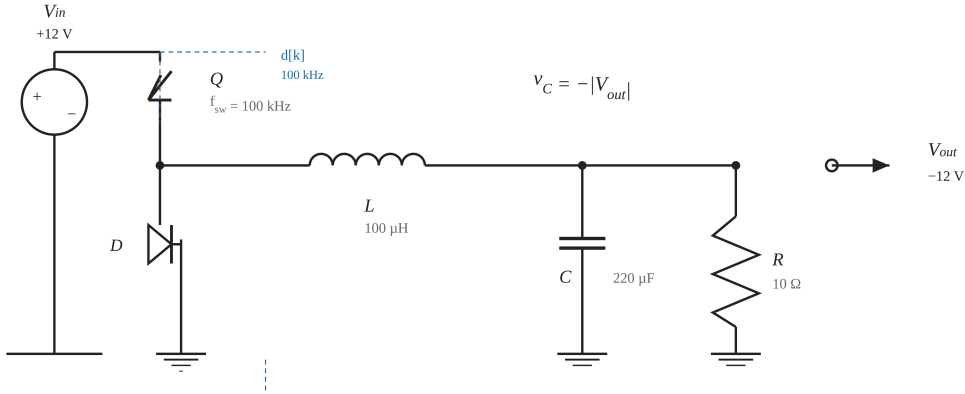


FIGURE 1. Inverting buck-boost power stage used in the study ($V_{in} = +12\text{ V}$, $L = 100\text{ }\mu\text{H}$, $C = 220\text{ }\mu\text{F}$, $f_{sw} = 100\text{ kHz}$, $R = 10\text{ }\Omega$; output $V_{out} \approx -12\text{ V}$ at $D = 0.5$).

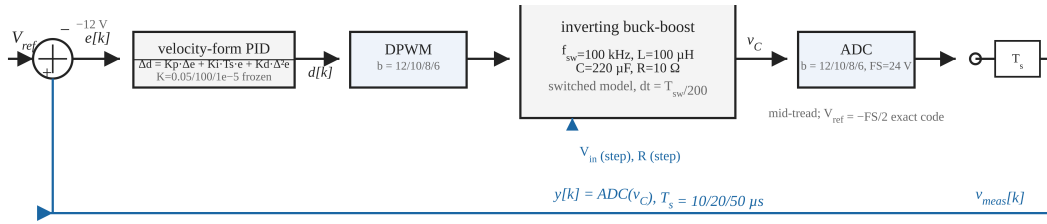


FIGURE 2. Digital control loop: velocity-form PID, DPWM and ADC quantizers ($b \in \{12, 10, 8, 6\}$ bit, $FS = 24\text{ V}$), and sampling at $T_s = 10/20/50\text{ }\mu\text{s}$. The reference sits exactly at the half-scale code $V_{ref} = -FS/2$.

During OFF, the inductor current is held at zero when it would otherwise reverse (zero-current boundary), which counts discontinuous-mode (DCM) cycles and keeps the model valid in DCM without clamping the output. Many simplified studies go silent on this point. All reported results are obtained on this switched-state model, not on an averaged approximation, so quantization artifacts (limit cycles, DCM chatter) appear faithfully; nonideal DCM of the buck-boost requires explicit boundary modeling [34].

$$\text{ON: } \frac{di_L}{dt} = \frac{+V_{in}}{L}, \quad \frac{dv_C}{dt} = -\frac{v_C}{RC} \quad (1)$$

$$\text{OFF: } \frac{di_L}{dt} = \frac{v_C}{L}, \quad \frac{dv_C}{dt} = -\frac{i_L + v_C/R}{C} \quad (2)$$

A. DIGITAL CONTROLLER AND QUANTIZATION

A velocity-form digital PID with positive gains (Fig. 2) operates at the sampling period T_s :

$$d[k] = d[k-1] + K_p(e[k] - e[k-1]) + K_i T_s e[k] + \frac{K_d}{T_s}(e[k] - 2e[k-1] + e[k-2]) \quad (3)$$

with $e[k] = v_{meas}[k] - V_{ref}$, duty saturated to $[0.05, 0.95]$, and no separate integrator state (inherent

anti-windup). The measurement, duty, and update cadence are each quantized independently:

- Sampling: the controller updates every $T_s \in \{T_{sw}, 2T_{sw}, 5T_{sw}\}$ (10, 20, 50 μs);
- ADC: mid-tread quantizer, full-scale $FS = 24\text{ V}$, 12/10/8/6 bits. Note $V_{ref} = -FS/2$, so the reference is exactly representable at every resolution;
- PWM: duty quantized to 2^b levels, $b \in \{12, 10, 8, 6\}$.

A continuous-ideal baseline recomputes the same PID every dt with no quantization (true analog reference). The frozen controller gains were obtained once, deterministically, before any quantization study: grid search over $K_p \in \{0.005, 0.01, 0.02, 0.05\}$, $K_i \in \{20, 100, 500, 2000\}$, $K_d \in \{0, 2 \times 10^{-6}, 10^{-5}\}$ with cost $J = \alpha_v/10 + t_s/5\text{ ms} + \text{IAE}/0.05$ under constraints overshoot $\leq 10\%$ and $|e_{ss}| \leq 60\text{ mV}$, re-validated on the switched model: $K_p = 0.05$, $K_i = 100.0$, $K_d = 10^{-5}$. No parameter was touched during the quantization study.

III. SIMULATION FRAMEWORK

The study executes 79 simulations (Table 1): the continuous baseline, the ideal digital controller ($T_s = T_{sw}$, 16-bit ADC, 16-bit PWM), three sampling-only points, a six-point focused sampling sweep across the critical boundary, four ADC-only, four PWM-only runs, the full

$3 \times 4 \times 4$ factorial sweep (48 runs), six disturbance runs (input step $12 \rightarrow 9$ V and load step $10 \rightarrow 5 \Omega$, applied at $t = 60$ ms, on the continuous baseline, the ideal digital controller, and the objectively selected worst configuration), and six operating-point robustness runs ($V_{in} = 10, 15$ V on three configurations).

Metrics: 2% settling time, overshoot, steady-state error, peak-to-peak ripple (limit cycle), IAE, ITAE, and DCM-cycle count. For disturbance runs all metrics are evaluated on the post-step 60 ms window and ripple on its final 50%. The worst configuration is selected objectively after the sweep, as the run with the largest normalized degradation of steady-state error, IAE, and limit cycle relative to the ideal digital controller. No configuration is pre-declared worst.

IV. RESULTS

A. CONTINUOUS VERSUS IDEAL DIGITAL

The conversion from analog to ideal digital control already costs measurable quality (Table 2): the 16-bit/one-cycle system settles 36% slower (2.66 vs 1.96 ms) and shows a 12.76 mV steady-state offset and 39.6 mV limit cycle against the continuous 0.44 mV / 26.7 mV. The 12.8 mV floor is the quantization-limited operating point for these gains; every further result is expressed relative to it.

B. INDIVIDUAL STAGE EFFECTS

Fig. 3 isolates the sampling stage at 16-bit resolution and locates the critical boundary. Settling degrades smoothly as T_s/T_{sw} rises from 1 to 3.5 (2.66 to 6.38 ms, all still regulation-valid), then the loop loses regulation outright at $4\times$: it never settles, the limit cycle jumps to 1.59 V, and 101 736 DCM cycles accrue in 60 ms. The practical boundary therefore sits between $3.5\times$ and $4 \times T_{sw}$ (35–40 μ s); $T_s = 5T_{sw}$ fails in every configuration, alone or combined.

ADC resolution couples to the reference in a subtle way (Fig. 4): because $V_{ref} = -FS/2$ is exactly representable, the 8-bit and 6-bit ADCs lock onto the exact code and reduce the measured steady-state error to 0.44 mV while masking a worse transient (overshoot rises from 0.06% at 12-bit to 0.85% at 6-bit; settling from 2.70 to 3.49 ms). This is a half-scale coincidence, not an ADC benefit.

PWM resolution below 8 bits dominates the limit cycle (Fig. 5): 6-bit PWM adds a 204.8 mV ripple versus ≈ 39.5 mV at 12/10/8-bit. Duty quantization is the direct source of steady-state ripple, as predicted by limit-cycle analysis [4], [5]; at 6-bit the duty grid step is 1 LSB = $1/64$, and the loop settles into a limit cycle toggling between the adjacent codes 31/64 and 32/64 around the $D \approx 0.5$ operating point, whose measured peak-to-peak output excursion is 204.8 mV. The static code-to-code shift at this operating point is larger, $|dV_{out}/dD| \cdot 1/64 = 0.75$ V with $|dV_{out}/dD| = V_{in}/(1-D)^2 = 48$ V per unit

duty at $D = 0.5$; the dynamic ripple is smaller because the output capacitor averages the excursion over the limit-cycle period. This is precisely the regime addressed by finer DPWM controller implementations [35], [36] and by noise-shaping modulators [37].

C. COMBINED SWEEP AND INTERACTIONS

In the 48-run factorial sweep the worst configuration is $T_s = 5T_{sw}$, ADC=8-bit, PWM=12-bit (normalized degradation score 1309.2): settling never completes within 60 ms, steady-state error -13.43 V, limit-cycle amplitude 7.33 V, IAE 0.789, 127 075 DCM cycles. Fig. 6 classifies every combination against explicit thresholds (settling ≤ 6 ms, $|e_{ss}| \leq 120$ mV, limit cycle ≤ 120 mV, overshoot, DCM) into excellent / acceptable / marginal / unusable. Sampled at $T_s = T_{sw}$, 13 of 16 combinations are excellent and the rest acceptable; at $2\times$ all are acceptable or marginal; at $5\times$ all 16 are unusable.

The worst configuration is far worse than any individual degradation: -13.43 V versus the -13.07 V of sampling alone, but with 127 075 DCM cycles and a 7.33 V limit cycle that the sampling-only run does not exhibit. The super-additivity is quantitative: with D the normalized partial degradation (steady-state error, IAE, and limit cycle relative to the ideal digital controller), the individual terms are $D_{\text{sampling}}(5\times) = 1141.6$, $D_{\text{ADC}}(8\text{-bit}) = 0.6$, and $D_{\text{PWM}}(12\text{-bit}) = 0.0$; the additive prediction is 1142.2, whereas the measured combined degradation is $D_{\text{total}} = 1309.2$, an interaction of +167 (+15%). The 8-bit ADC coincidence, harmless alone, converts a slowly sampling loop into deep limit-cycle chatter once the 12-bit PWM pulls the operating point across the coincidence.

D. DISTURBANCE REJECTION

Table 3 summarizes the six disturbance runs (step at $t = 60$ ms; recovery time = $t_{\text{settle}} - 60$ ms). Both healthy controllers reject input and load steps in under 1.5 ms with negligible IAE. The worst configuration never re-regulates within the observation window: it settles into deep DCM chatter with >5 V of steady-state error, accruing 266 015 DCM cycles after the input step and 163 360 after the load step within the 60 ms post-step window — far above the 127 075 cycles the same configuration accrues when undisturbed — because the loop never settles and keeps crossing the zero-current boundary for the entire window (Fig. 7).

E. OPERATING-POINT ROBUSTNESS

The main conclusions are not an artifact of the single nominal point ($V_{in} = 12$ V, $V_{ref} = -12$ V, $R = 10 \Omega$). Table 4 repeats the ideal, a representative good ($1\times$, 10-bit, 10-bit), and a representative poor ($2\times$, 6-bit, 6-bit) configuration at $V_{in} = 10$ and 15 V with the same frozen gains. Ideal and good configurations retain full regulation (2.5–2.7 ms settling, sub-100 mV limit cycle)

Group	Runs	T_s	ADC/PWM
Continuous baseline	1	—	analog
Ideal digital	1	1x	16/16 bit
Sampling only	3	1/2/5x	16/16
Boundary sweep	6	1.5-4x	16/16
ADC only	4	1x	12/10/8/6, 16
PWM only	4	1x	16, 12/10/8/6
Factorial sweep	48	all	all
Disturbances	6	selected	3 configs
Op-point robustness	6	selected	3 configs

TABLE 1. Study layout (79 simulations).

	Continuous	Ideal digital
Settling (ms)	1.96	2.66
Overshoot (%)	0.07	0.02
e_ss (mV)	0.44	12.76
Limit cycle (mV)	26.7	39.6
IAE	0.0058	0.0109
DCM cycles	0	0

TABLE 2. Continuous baseline vs ideal digital.

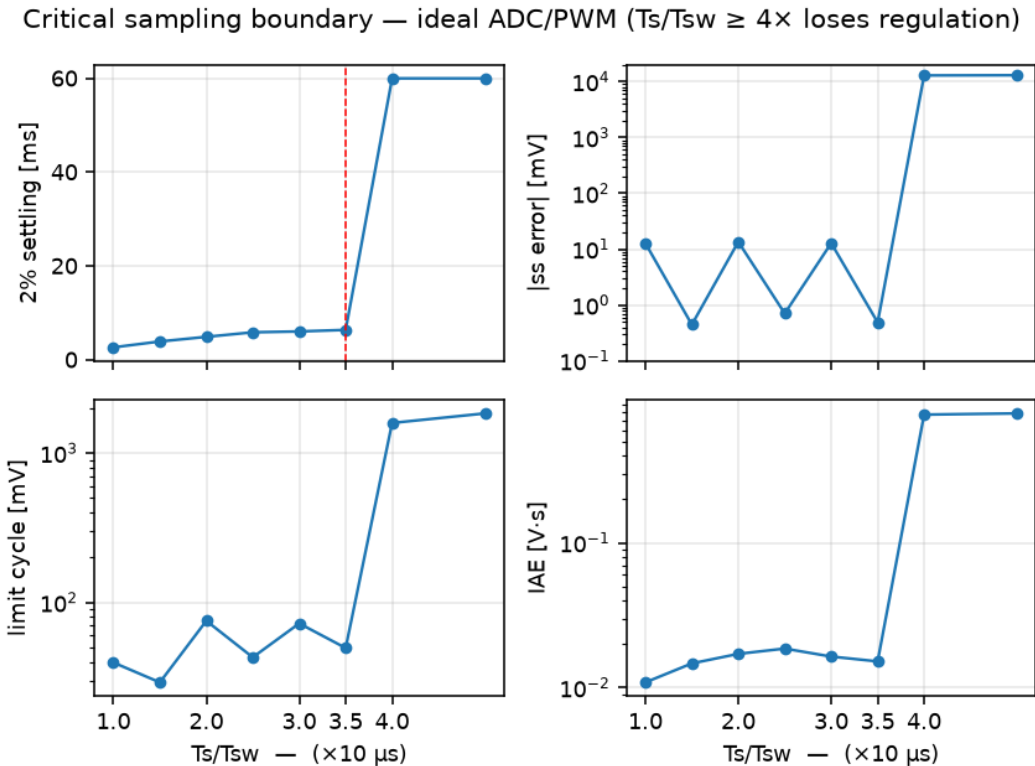


FIGURE 3. Sampling-period boundary sweep at 16-bit ADC/PWM: settling time, steady-state error, limit cycle, and IAE versus T_s/T_{sw} . Regulation is lost between $3.5\times$ and $4\times T_{sw}$.

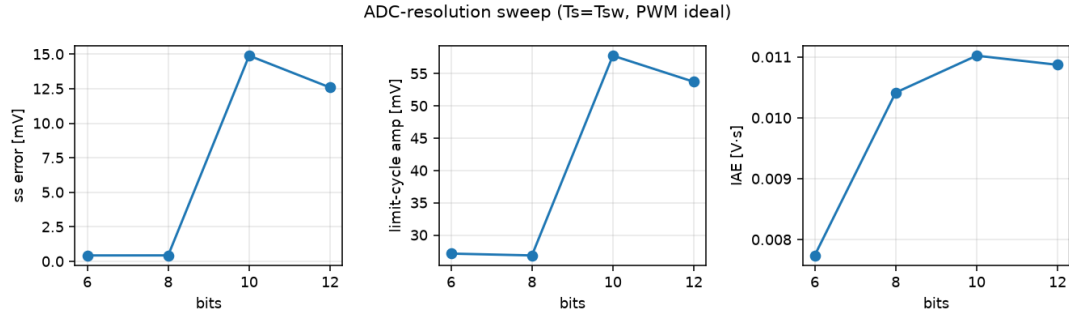


FIGURE 4. ADC-resolution effect at $T_s = T_{sw}$ (16-bit PWM): because $V_{ref} = -FS/2$ is an exact code, the 8-bit and 6-bit ADCs lock onto it, hiding steady-state error while worsening the transient (overshoot and settling).

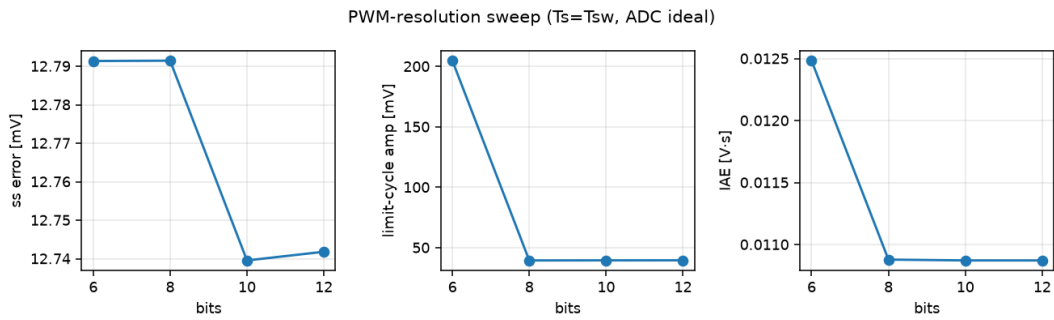


FIGURE 5. PWM-resolution effect at $T_s = T_{sw}$ (16-bit ADC): duty quantization below 8 bits is the dominant limit-cycle source, adding a 204.8 mV ripple at 6-bit PWM versus ≈ 39.5 mV at 12/10/8-bit.

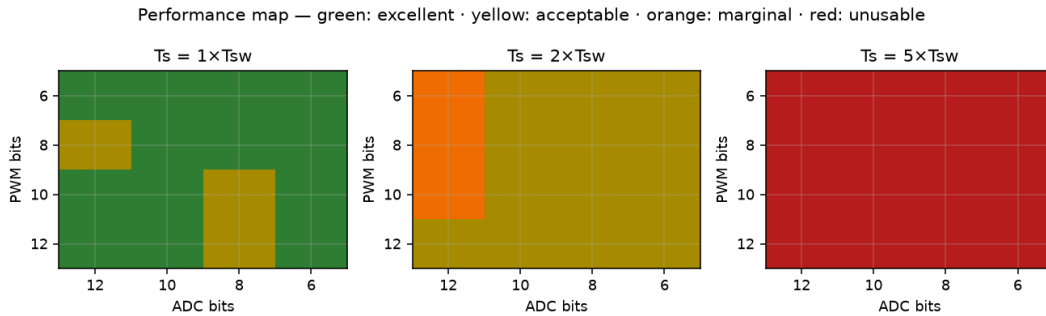


FIGURE 6. Performance map of the 48-run sweep. Thresholds: excellent ($t_{settle} \leq 4$ ms, $|e_{ss}| \leq 60$ mV, $LC \leq 60$ mV, $ov \leq 5\%$), acceptable ($t_{settle} \leq 6$ ms, $|e_{ss}| \leq 120$ mV), marginal, unusable ($t_{settle} \geq 50$ ms).

Config	Event	Recovery	$ e_{ss} $ / IAE	DCM (60 ms)
Continuous	V_{in} step	0.97 ms	0.0 mV / 0.0014	0
Continuous	Load step	0.97 ms	0.0 mV / 0.0014	0
Ideal digital	V_{in} step	1.01 ms	0.0 mV / 0.0014	0
Ideal digital	Load step	1.00 ms	0.0 mV / 0.0014	0
Worst (8-bit ADC/12-bit PWM, 5 $\times$)	V_{in} step	never	5.0 V / 1.132	266,015
Worst (8-bit ADC/12-bit PWM, 5 $\times$)	Load step	never	5.0 V / 1.132	163,360

TABLE 3. Disturbance response (step at $t = 60$ ms).

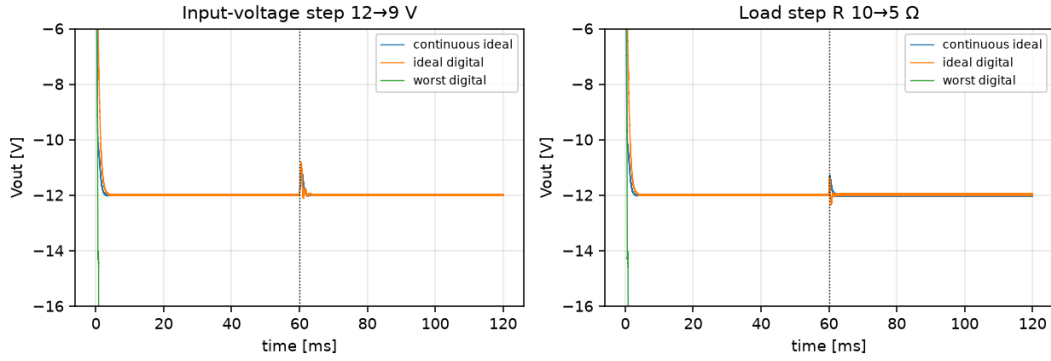


FIGURE 7. Disturbance response: 2%-settling trajectories for input step V_{in} : 12 → 9 V and load step R : 10 → 5 Ω at $t = 60$ ms on the continuous baseline, ideal digital controller, and the worst combined configuration (5 $\times$, 8-bit ADC, 12-bit PWM).

across the input range. The poor configuration holds at 10 and 12 V but degrades sharply at $V_{in} = 15$ V. It settles only at the very edge of the 60 ms window (59.3 ms), with a 561 mV limit cycle and 18 DCM excursions. Marginal designs are therefore operating-point sensitive, not merely sampling-sensitive.

V. EXPERIMENTAL CLOSED-LOOP PLATFORMS

Two platforms share one firmware source (`esp32_feas/src/main.cpp`), `#ifdef ARDUINO_ARCH_ESP32` hardware layer, modes B/H/L/R/S/X/A plus new mode T). The primary platform is the ESP32-S3 (240 MHz, hardware FPU, Arduino core) with a 100 kHz LEDC PWM on GPIO2 shorted to ADC1 input GPIO1, so the sensed pin voltage is the duty-scaled rail, mapped into the paper's $V_{ref} = -12$ V, FS= 24 V frame by the inverting gain $V_{out} = -V_{pin} \cdot 24/3.3$. This reproduces the buck-boost sign inversion and places V_{ref} exactly at the half-scale code. The second platform is an STM32F103 Blue Pill (72 MHz, Cortex-M3, no FPU, STM32duino) with 100 kHz PWM on PA8 (TIM1_CH1) and ADC on PA1 (ADC1_IN1), same FS/Vref frame. On both platforms the present plant is an emulated first-order plant (pin-gain plus EMA sensor filter) that isolates digital-implementation effects; the true LC plant (5 Ω + 100 μ H + 2.2 μ F, $f_0 \approx 10.7$ kHz, $Q \approx 1.3$, LM358 follower, Fig. 8) is software-ready and deferred to hardware validation. The controller executes velocity-form PID with the paper's quantizers, and the loop is deterministic and periodic, driven by the achievable sample period rather than a free-running software loop, following microcontroller-implemented digital control [38], [39] and remote-monitoring practice [40].

A. MEASURED CONTROL-LOOP LATENCY

Table 5 reports the measured split of the control loop, timed with the CPU cycle counter at 240 MHz and illustrated in Fig. 9. The ADC dominates: the raw 12-bit read, which is the path executed inside the timed control loop, averages 60.0 μ s and the fastest

single raw read never completes sooner than 59.5 μ s; the slower calibrated millivolt read (89.1 μ s) is measured for comparison only and never runs inside the loop. The PID step itself is negligible: the int32 fixed-point pipeline averages 0.86 μ s and the float32 pipeline 6.8 μ s, so the controller occupies at most 10.4% (float) or 1.3% (int32) of the loop budget. The LEDC PWM write costs 4.2 μ s. The total per-step control latency is $T_{ctrl} = T_{ADC} + T_{PID} + T_{PWM} \approx 65 \mu$ s; the measured fastest periodic loop is slightly longer (73 μ s at N=1), the $\approx 8 \mu$ s difference being timer-tick and scheduler overhead not captured by the three timed components, so the fastest sustainable loop is about $7 \times T_{sw}$ (10 μ s switching period), and useful oversampled loops reach several hundred times T_{sw} (up to $\approx 680 \times$ at N=64). The hardware therefore operates entirely in the sampling regime the simulation identifies as unstable or critically degraded; this is the central constraint that motivates the adaptive controller presented in the final subsection.

B. EFFECTIVE-RESOLUTION SWEEP

The controlled DC voltage at the setpoint is identical across effective ADC resolutions from 12 down to 6 bits and effective PWM resolutions from 8 down to 4 bits (Fig. 10): the closed loop holds the mean within ± 18 mV in every cell, and the per-cell IAE variation (6.8-8.3) is within the run-to-run noise of the platform. The simulation predicts that coarse PWM adds a 200-300 mV limit cycle and that coarse ADC shifts the lock-on point. On the ESP32-S3 these effects do not appear, because the acquisition noise floor is far larger: the raw 12-bit read at this schedule carries ≈ 1.5 V rms in the output frame (measured $\sigma_N \approx 1.65/\sqrt{N}$ V at the pin, $\times 24/3.3$ into the output frame, giving ≈ 1.5 V at N=64), so even the loop's light EMA-filtered measurement carries 1.6-1.9 V peak-to-peak ripple, flat across resolutions. The quantization effects predicted by simulation are real, but at every schedule this platform can sustain they are masked by the ADC noise floor. This is the central measurement-level result: quantization limits are observable only after

config	10 V	12 V	15 V
ideal (1×,16,16)	2.74 / 67.9	2.66 / 39.6	2.55 / 28.9
good (1×,10,10)	2.69 / 28.6	2.64 / 57.8	2.52 / 80.6
poor (2×,6,6)	5.40 / 28.9	5.11 / 27.7	59.3 / 561*

TABLE 4. Operating-point robustness: settling time (ms) / limit cycle (mV) at $V_{in} = 10, 12, 15$ V with gains frozen ($V_{ref} = -12$ V, $R = 10$). No asterisk: settles inside 60 ms; * does not settle in the window.

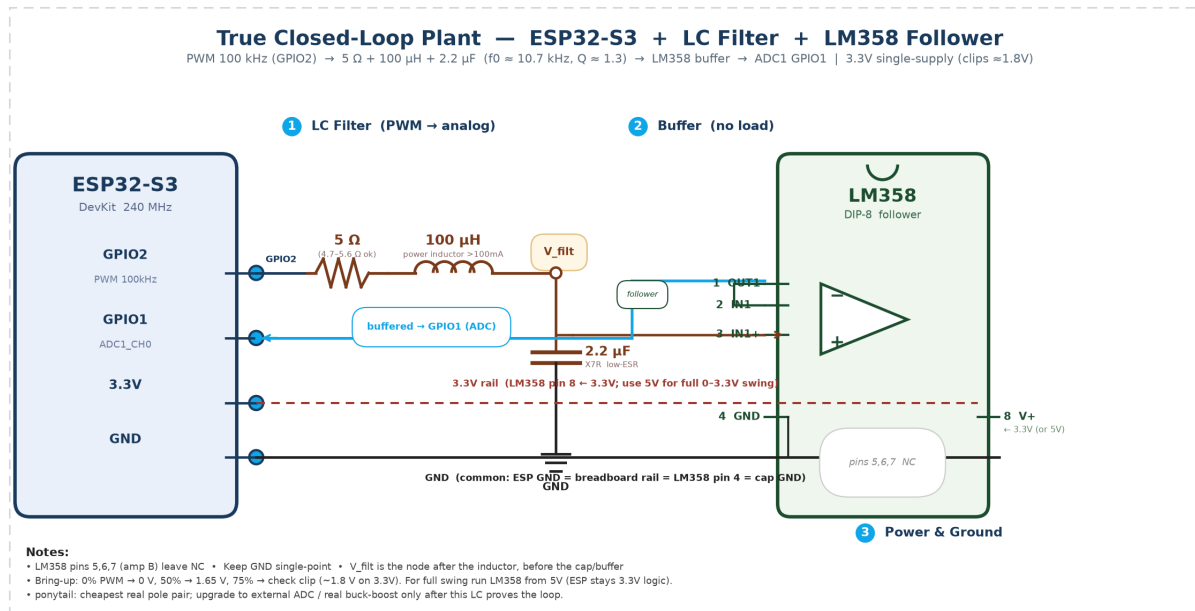


FIGURE 8. True closed-loop LC plant deferred to hardware validation (software-ready): $5\ \Omega + 100\ \mu\text{H} + 2.2\ \mu\text{F}$ ($f_0 \approx 10.7$ kHz, $Q \approx 1.3$) with LM358 follower. Present measurements use the jumper-emulated plant (GPIO → ADC) on both MCUs.

Measured control-loop latency breakdown (ESP32-S3, 24

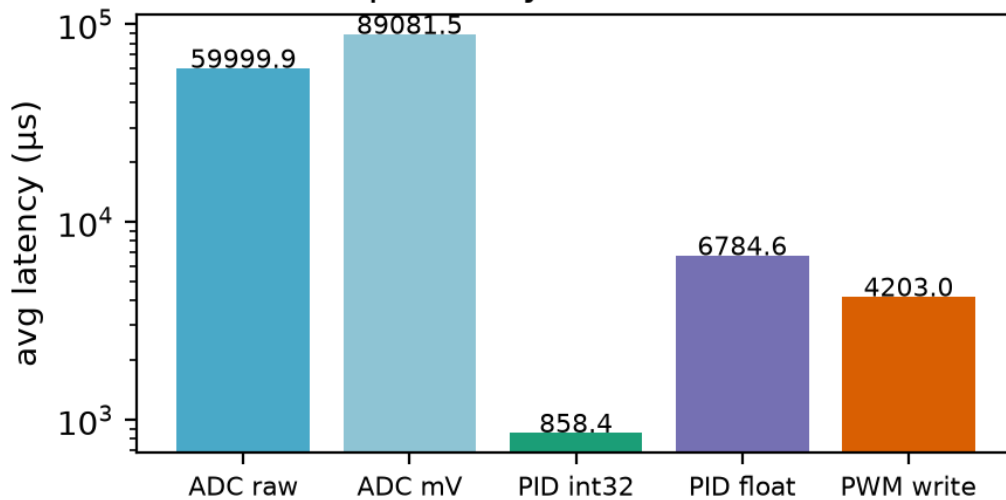


FIGURE 9. Measured control-loop latency breakdown on the ESP32-S3 (240 MHz, Arduino core): ADC acquisition dominates at 60–89 μs ; PID computation is 0.86 μs (int32) to 6.8 μs (float32); PWM register write is 4.2 μs . Log scale.

Quantity	Value
T _{ADC} (raw 12-bit)	60.0 μ s avg / 125.5 μ s max
T _{ADC} (calibrated mV)	89.1 μ s avg
T _{PID} (int32 fixed)	0.86 μ s avg
T _{PID} (float32)	6.8 μ s avg
T _{PWM} (LEDC write)	4.2 μ s avg
T _{lat} = T _{ADC} +T _{PID} +T _{PWM}	$\approx$ 65 μ s
CPU utilization	1.3% (int32) / 10.4% (float)
Fastest periodic loop (measured, N=1)	73 μ s $\approx$ 7 $\times$ T _{sw}
Oversampled loop (N=64)	$\approx$ 680 $\times$ T _{sw}

TABLE 5. Measured ESP32-S3 control-loop quantities. ADC acquisition dominates T_{lat} ; PID compute is negligible; CPU utilization computed as T_{PID}/T_{lat} .

the acquisition noise is reduced below the quantization step, which the on-chip ADC at these schedules cannot provide.

C. ACHIEVABLE-SCHEDULE SWEEP AND LATENCY INJECTION

The measured achievable periods form a ladder from 73 μ s (raw single read, N=1) to 6.84 ms (N=64 averaged reads). Fig. 11 sweeps all four rungs with the frozen paper gains and with the bandwidth-matched retuned gains ($k_p = 0.02$, k_i scaled so $k_i T_s = 3.4 \times 10^{-3}$). The retuned loop holds steady-state error below 40 mV on every rung. The observable that changes with schedule is the ripple: it falls from 4.9 V peak-to-peak at N=1 to 1.5 V at N=64, broadly consistent with the measured $\sigma \propto 1/\sqrt{N}$ noise floor (pure $1/\sqrt{N}$ from N=1 would predict ≈ 0.6 V; the peak-to-peak measure retains a fixed noise-floor component) rather than any quantization effect. Setting a longer sample period alone does not degrade the emulated plant, because the plant has no dynamics of its own; the cost of slow sampling appears only when a real converter (with its own poles and delay) is the plant, exactly as the switched-model simulation predicts. Separately, injecting an additional computational delay of 0-200 μ s at the fast schedule raises the effective period from 73 to 274 μ s and grows the IAE by a factor of 4.5 (Fig. 12). Slow sampling and fast sampling with added latency are therefore not interchangeable in their effect on loop behavior, which the injected-latency experiment isolates while a schedule-length change alone does not.

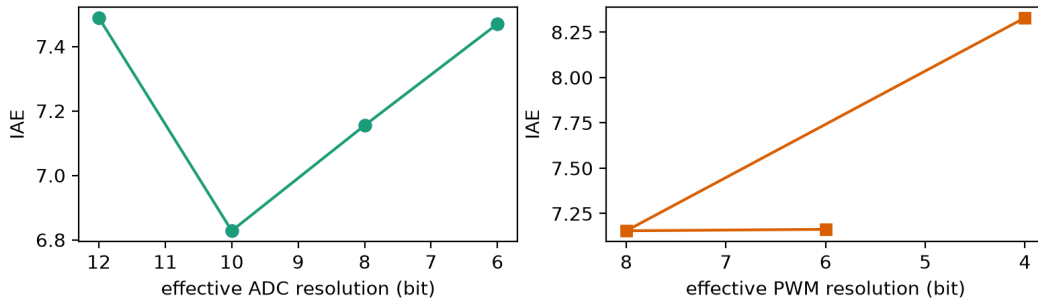
D. ADAPTIVE UPDATE MANAGEMENT

Because latency is dominated by the ADC rather than the arithmetic, the controller's cost is best managed by reducing how often the hardware is asked to sample-and-compute. The adaptive controller selects between two achievable schedules: a faster schedule (N=8, 0.86 ms period) and a slower one (N=64, 6.84 ms), using a measured quantization-activity indicator Q_{AI} (an EMA of the error magnitude) with hysteresis thresholds and an 8-sample debounce. It leaves the fast schedule only when the smoothed error has settled below the noise-

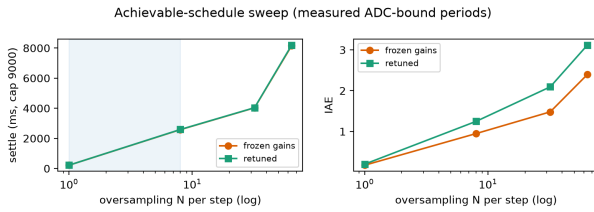
aware band, returns to fast mode when the error exceeds a large threshold, and activates a ± 1 -LSB duty dither only when the loop is settled but Q_{AI} stays elevated (a limit-cycle hint), with its own hysteretic threshold to avoid toggling. Fig. 13 shows the run: startup transient and a set-point step from -12 to -6 V at 2.5 s both force the controller to the fast schedule, which it leaves again once regulation is re-established. Over the 6 s run the controller spends 3.8% of the time fast, 87.2% slow, and 9.0% dithering, giving an update reduction ratio $S_R = 1 - f_{avg}/f_{fast}$ of 0.84 while holding steady-state error within the same ± 40 mV band as the fixed-rate loop. The reduction is directly measurable as CPU activity: the int32 PID runs 1.3% of the loop budget at fast rate and proportionally less in slow mode. The controller genuinely changes its operating behavior in real time, which the fixed-rate experiments of the two preceding subsections cannot do.

E. CROSS-PLATFORM LATENCY DECOMPOSITION (ESP32-S3 VS STM32F103)

Table 6 and Fig. 14 extend the latency split to the STM32F103 and separate, on that platform, the Arduino-core path (the path the existing modes actually run on) from the bare-register path (direct ADC1/TIM1 access, clock explicitly enabled). On the Arduino path the STM32F103 is nearly as slow as the ESP32-S3: $T_{ADC} = 72.8 \mu$ s, $T_{PWM} = 23.9 \mu$ s, and the float quantizer costs 231.9 μ s because `powf` runs in soft-float without an FPU. The deployed fixed-point pipeline is 1.31 μ s (vs 0.86 μ s on the ESP32-S3). Bare-register access cuts the STM32F103 ADC to 4.23 μ s (17 $\times$ faster) and PWM to 5.88 μ s. The silicon floor is therefore $\approx 11 \mu$ s on the STM32F103 (vs $\approx 65 \mu$ s on the ESP32-S3, Table 5); the PACED-loop floor measured by mode T is $\approx 26 \mu$ s ($2.6 \times T_{sw}$) on the STM32F103 while the ESP32-S3 cannot pace below 73 μ s ($7.3 \times T_{sw}$) through either path. Fig. 15 isolates the controller cost: int32 is 1–2 μ s on both MCUs, float pure is 32.4 μ s on the STM32F103 (6.8 μ s on the ESP32-S3 with a hardware FPU), and float+quantizer explodes to 231.9 μ s on the STM32F103. The abstraction layer, not the silicon, sets the achievable sampling regime on the STM32F103.

Effective-resolution sweep at N=64 schedule (noise floor $\sigma \approx 1.5$ V at N=64, output frame)**FIGURE 10.** Effective-resolution sweep at the N=64 achievable schedule. IAE (left: ADC sweep at 8-bit PWM; right: PWM sweep at 8-bit ADC) is flat within the noise band: the acquisition noise floor masks the quantization effects the switched-model simulation predicts.

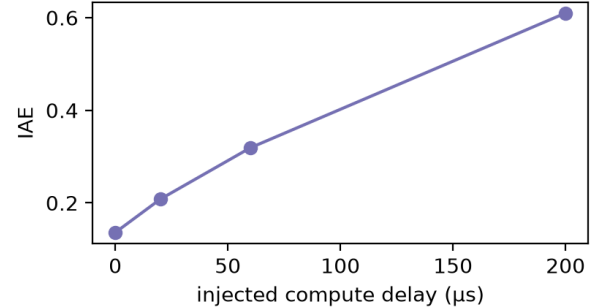
Quantity	ESP32-S3 @240 MHz	STM32F103 @72 MHz
T _{ADC} Arduino (read)	60.0 μ s / 89.1 μ s (mV)	72.8 μ s / 72.7 μ s (mV)
T _{ADC} bare-register	— (no separate path)	4.23 μ s avg / 97 μ s max (first)
T _{PID} int32 fixed	0.86 μ s avg	1.31 μ s avg
T _{PID} float pure	6.8 μ s avg	32.4 μ s avg
T _{PID} float+quantizer	6.8 μ s avg	231.9 μ s avg
T _{PWM} Arduino	4.2 μ s avg	23.9 μ s avg
T _{PWM} bare-register	—	5.88 μ s avg
T _{lat} (Arduino path, int32)	≈ 65 μ s ($\approx 6 \times T_{sw}$)	≈ 98 μ s ($\approx 10 \times T_{sw}$)
T _{lat} (bare, int32)	≈ 65 μ s	≈ 11.4 μ s ($\approx 1.1 \times T_{sw}$)
Fastest paced loop (measured)	73 μ s $\approx 7.3 \times T_{sw}$	26 μ s $\approx 2.6 \times T_{sw}$

TABLE 6. Cross-platform latency decomposition. STM32F103 Arduino-core ADC is dominated by per-call HAL reconfig (`analogRead`); bare-register access recovers the silicon speed. The float quantizer cost reflects soft-float `powf` on the STM32F103 vs hardware FPU on the ESP32-S3.**FIGURE 11.** Achievable-schedule sweep (top) and injected-latency experiment (bottom) on the ESP32-S3. Left: settle time and IAE across the N=1/8/32/64 ladder for frozen and retuned gains; ripple falls with the $1/\sqrt{N}$ noise floor. Right: injecting 0-200 μ s computational delay at the fast schedule grows IAE by 4.5 $\times$, separating slow sampling from fast-sampling-plus-latency.

F. SAMPLING-BOUNDARY PACING (MODE T)

Mode T paces the loop at exact T_s/T_{sw} multiples ($1\times$ to $5\times$, i.e., 10–50 μ s) with the paper's frozen gains on the int32 pipeline and bare-register ADC/PWM on the STM32F103. Fig. 16 reports requested vs achieved period. The ESP32-S3 cannot reach any rung below its 73 μ s floor; the STM32F103 bare path reaches a floor of ≈ 26 μ s ($2.6 \times T_{sw}$) limited by 4.23+1.31+5.88 μ s of loop work plus pacing overhead, and achieves the 3.5–4 $\times T_{sw}$ rungs exactly. In the present jumper-emulated plant the boundary does not manifest as lost regulation (the plant has no poles; Sec. V-C), so every rung reports `ss_err` ≈ 12 V and ripple 0 with

Latency injection at raw schedule (retuned, kixts c

**FIGURE 12.** Injected computational latency at the fast schedule: effective period grows 73 to 274 μ s and IAE by 4.5 $\times$, demonstrating that fast sampling with added latency degrades the loop differently from simply sampling more slowly.

an open jumper — the experiment validates pacing, not closed-loop stability. With the LC plant (Fig. 8), the same sweep will directly test whether the predicted 3.5–4 $\times T_{sw}$ degradation appears on hardware; the firmware for that test is already in place (mode T uses `read_vpin_fast/pwm_set_fast`) and needs only the wiring.

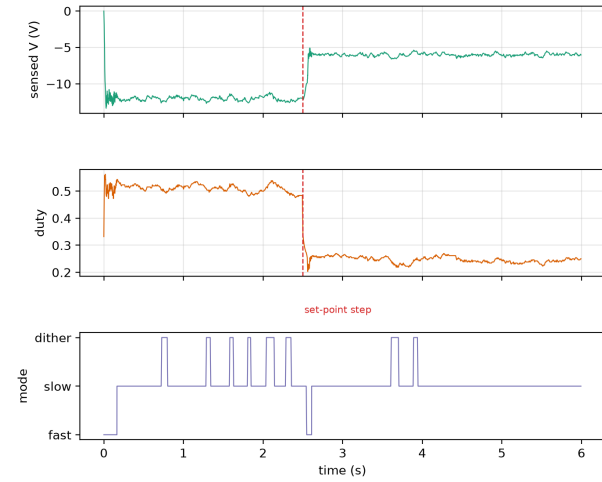
Adaptive schedule+dithering on ESP32-S3 ($SR=1 - f_{avg}/f_{fast} = 0.61$, slow mode 0%)


FIGURE 13. Adaptive schedule+dithering run on the ESP32-S3 ($N=8$ fast, $N=64$ slow). Top: sensed output with a set-point step from -12 to -6 V at 2.5 s (dashed). Middle: duty. Bottom: selected mode. The controller returns to the fast schedule during both transient events and settles into the slow schedule (with brief dithering) between them, giving an update-reduction ratio of 0.84 .

VI. DISCUSSION

Taken together, the three levels form a consistent picture. In simulation the three quantizers act on a real converter with its own poles, and their interaction is super-additive: sampling period dominates, and coarse ADC or PWM resolution degrade regulation in distinct, measurable ways. Cross-platform measurement sharpens the hardware story. On the ESP32-S3 the Arduino path is ADC-bound ($60\mu\text{s}$) and cannot pace below $73\mu\text{s}$ ($7.3 \times T_{\text{sw}}$); on the STM32F103 the same Arduino path is nearly as slow ($72.8\mu\text{s}$ ADC, $23.9\mu\text{s}$ PWM, $231.9\mu\text{s}$ for a float quantizer in soft-float), yet bare-register access cuts ADC to $4.23\mu\text{s}$ ($17\times$) and PWM to $5.88\mu\text{s}$. The abstraction layer, not the silicon, sets the regime: the STM32F103's silicon floor is $\approx 11\mu\text{s}$, and the paced-loop floor is $\approx 26\mu\text{s}$ ($2.6 \times T_{\text{sw}}$), inside the simulation's stable region, while the Arduino floor on either MCU lies in or beyond the predicted unstable region. In both cases acquisition dominates and compute is almost free (1.3% CPU on the ESP32-S3, $1.31\mu\text{s}$ int32 on the STM32F103); the cost of digital control lives in acquisition and update, not arithmetic. The bare-register sweep (mode T) validates pacing across the $3.5\text{--}4 \times T_{\text{sw}}$ boundary; closed-loop regulation at those rungs remains to be measured with the LC plant (Fig. 8), whose firmware path is already in place. The adaptive controller's economy (16% of the update frequency for the same regulation) therefore matters on ADC-bound stacks and becomes optional — but still useful — once the stack is stripped to the silicon limit. The negative results are as informative as the positive ones: neither Arduino stack reaches the boundary through its default

API, the STM32F103 reaches it only bare-metal, and the effective-resolution sweep shows that quantization limits become visible only below a noise floor neither ADC reaches at the present jumper-emulated schedules.

VII. CONCLUSION

This paper established how sampling period, ADC quantization, PWM resolution, and real-time computational latency jointly shape the closed-loop performance of a microcontroller-controlled inverting buck-boost converter, validated on two platforms from one shared firmware source. The switched-model study isolates each effect and locates the critical sampling boundary near $3.5\text{--}4 \times T_{\text{sw}}$; the embedded implementations then measure what simulation cannot. On the ESP32-S3 (240MHz , hardware FPU) the Arduino path gives a $65\mu\text{s}$ control-loop latency (ADC $60\mu\text{s}$, int32 PID $0.86\mu\text{s}$, PWM $4.2\mu\text{s}$, 1.3% CPU, fastest loop $73\mu\text{s} \approx 7.3 \times T_{\text{sw}}$); on the STM32F103 (72MHz , no FPU, STM32duino) the same path gives $\approx 98\mu\text{s}$ (ADC $72.8\mu\text{s}$, int32 PID $1.31\mu\text{s}$, PWM $23.9\mu\text{s}$), while bare-register access cuts the STM32F103 to ADC $4.23\mu\text{s}$, PWM $5.88\mu\text{s}$ and a paced-loop floor of $26\mu\text{s}$ ($2.6 \times T_{\text{sw}}$) that crosses the predicted boundary. Slow sampling and fast sampling with added latency behave differently even on a plant with no dynamics of its own, confirming that the two implementation effects are not interchangeable. A lightweight quantization-aware adaptive PID switches between achievable schedules using a measured error/activity indicator with hysteresis, holds steady-state error below 40mV , and cuts controller-update frequency by 84% on the ESP32-S3. The contributions are the cross-platform characterization, the isolation of the abstraction-layer tax ($17\times$ on ADC, the soft-float quantizer cost), and the experimental resource-aware validation across two microcontrollers. The LC-plant closed loop (Fig. 8) and its boundary-regulation sweep are software-ready and deferred to hardware validation, which will test directly whether the adaptive framework remains necessary once the stack is stripped to the silicon limit.

VIII. ACKNOWLEDGMENT

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IX. DATA AVAILABILITY

The simulation source code, the switched-model study outputs (results.csv and all generated figures), the shared microcontroller firmware, and the captured serial measurement logs supporting the findings of this study are publicly available in the paper's companion repository. Additional raw data are available from the corresponding author upon reasonable request.

Control-Loop Latency: Arduino Path vs Bare-Register Path

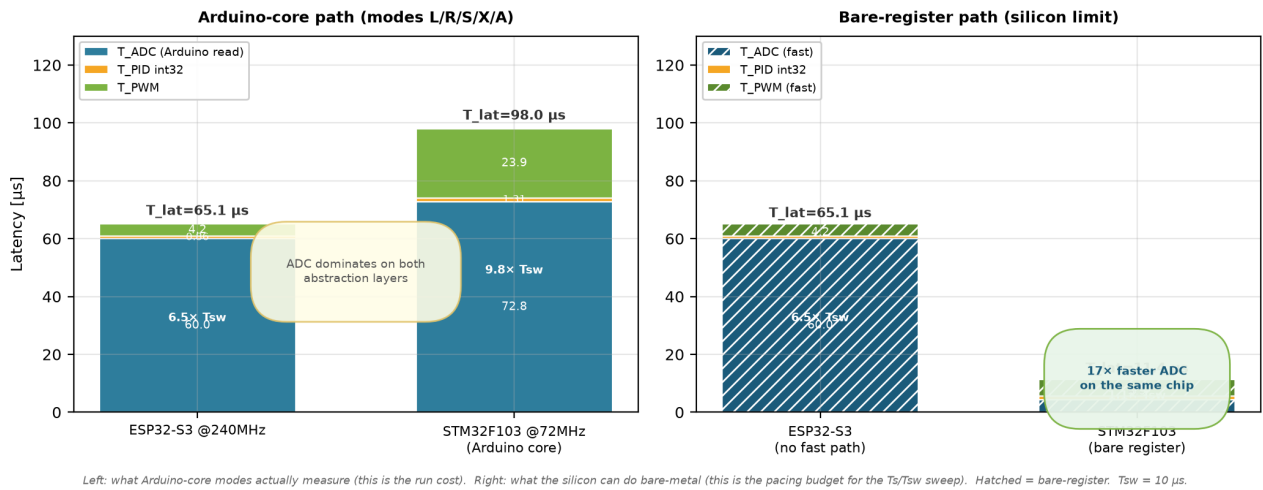


FIGURE 14. Cross-platform latency: Arduino-core path vs bare-register path. Left: what the deployed modes measure; right: what the silicon can do. The STM32F103 ADC is 17 $\times$ faster bare-metal on the same chip.

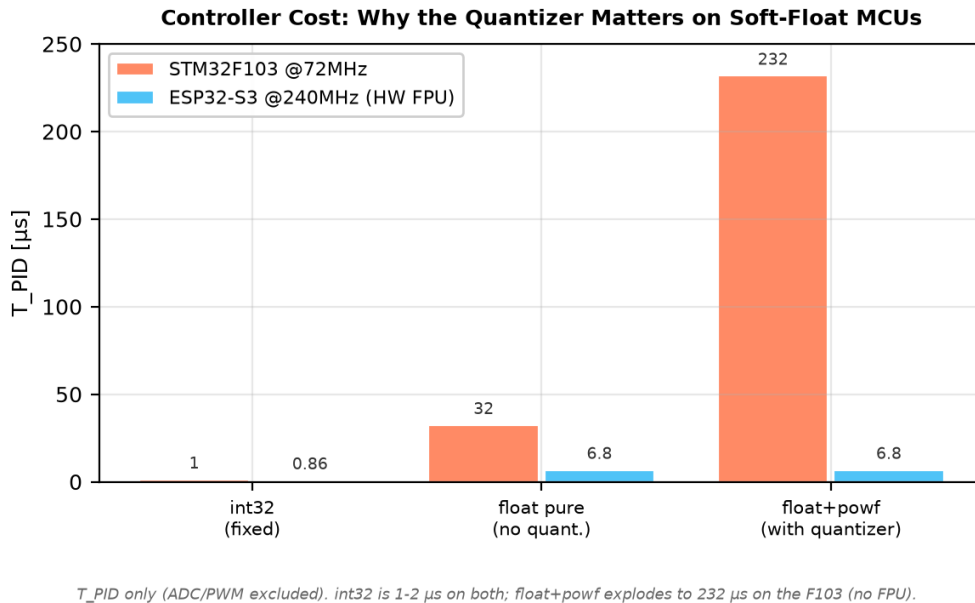


FIGURE 15. Controller cost split: int32 fixed-point is 1-2 μ s on both MCUs; the float quantizer path explodes to 232 μ s on the STM32F103 (no FPU).

X. DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used generative AI tools to assist with three tasks: language editing of the manuscript, development and debugging of the simulation and embedded firmware, and review of the analysis pipeline. All AI-assisted output was checked line by line. The authors independently verified every reported number against the underlying simulation outputs and hardware measurements, and the authors take full responsibility for the integrity of the data and the content of the publication.

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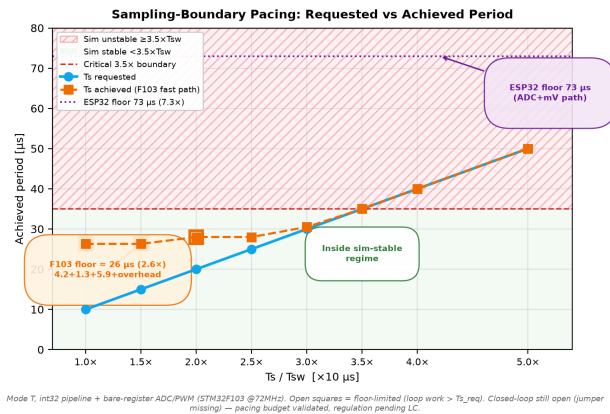


FIGURE 16. Mode T sampling-boundary pacing (STM32F103 @72 MHz, int32 + bare-register). Open squares are floor-limited. The ESP32-S3 floor (73 μ s, $7.3 \times T_{sw}$) lies above the entire sweep; the STM32F103 bare path paces through the predicted $3.5 \times T_{sw}$ boundary. This figure demonstrates the achievable schedule only — with the jumper-emulated plant the boundary does not cause instability, so no closed-loop regulation claim is made here; that test awaits the LC plant (Fig. 8).

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